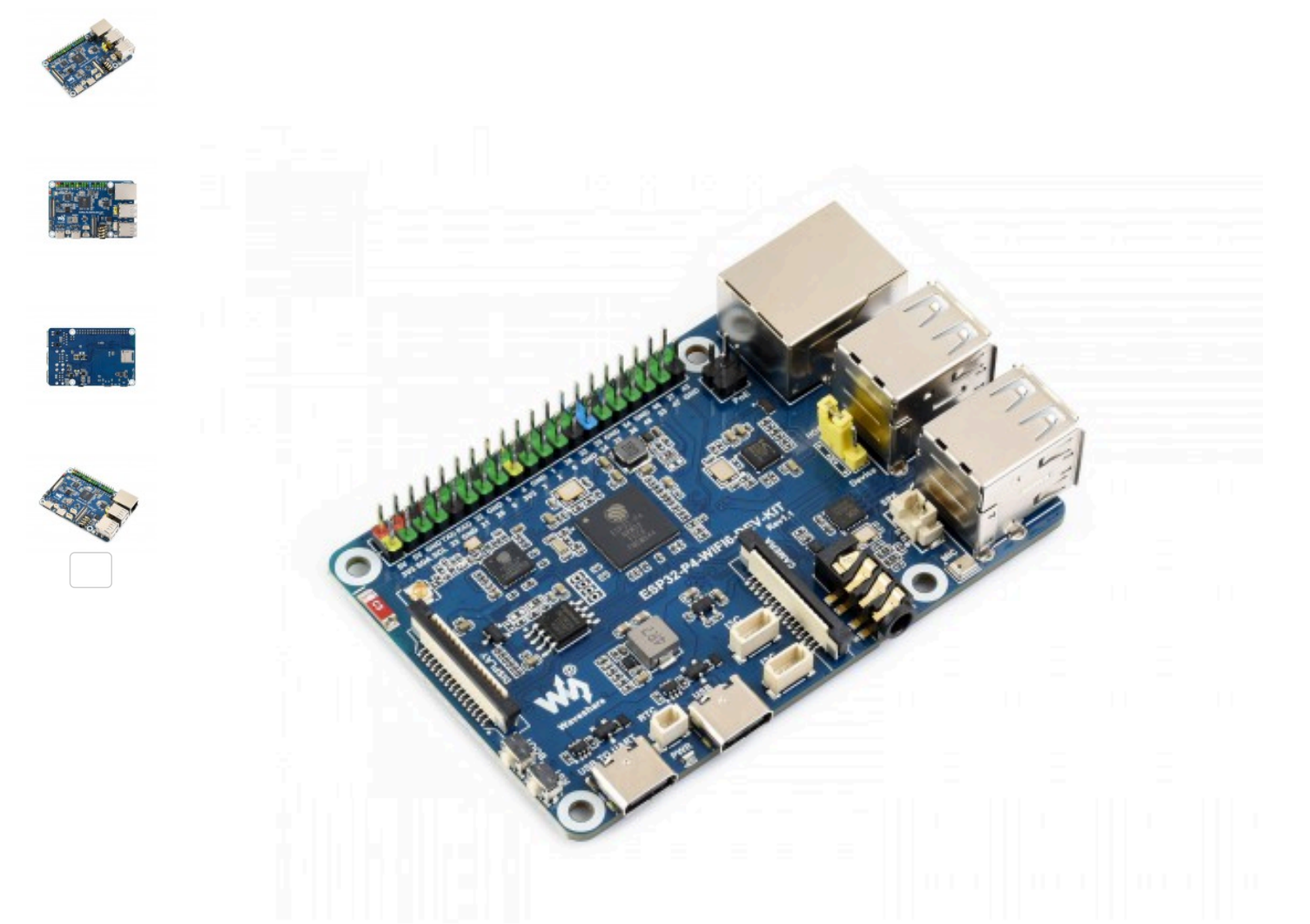




WiFi, Bluetooth, Ethernet, Arduino® Compatible, ESP32-P4, AI Chat, LCD (MIPI)

ESP32-P4-WIFI6-DEV-KIT High-Performance Development Board, Based On ESP32-P4 and ESP32-C6, Supports Wi-Fi 6 And Bluetooth 5 / BLE 5

SKU 32054
Part No. ESP32-P4-WIFI6-DEV-KIT
Brand Waveshare

\$18.99

1

ADD TO CART

Price

Chat
Quantity

\$18.99	1
\$18.39	2
\$18.09	3
\$17.97	4+

Version Options

- ESP32-P4-WIFI6-DEV-KIT Basic Kit
- ESP32-P4-WIFI6-DEV-KIT-A
- ESP32-P4-WIFI6-DEV-KIT-B
- ESP32-P4-WIFI6-DEV-KIT-C

DescriptionPackage ContentWiki

ESP32-P4-WIFI6-DEV-KIT High-Performance Development Board, Based On ESP32-P4 And ESP32-C6, Supports Wi-Fi 6 And Bluetooth 5 / BLE 5

ESP32-P4-WIFI6-DEV-KIT

Development Board

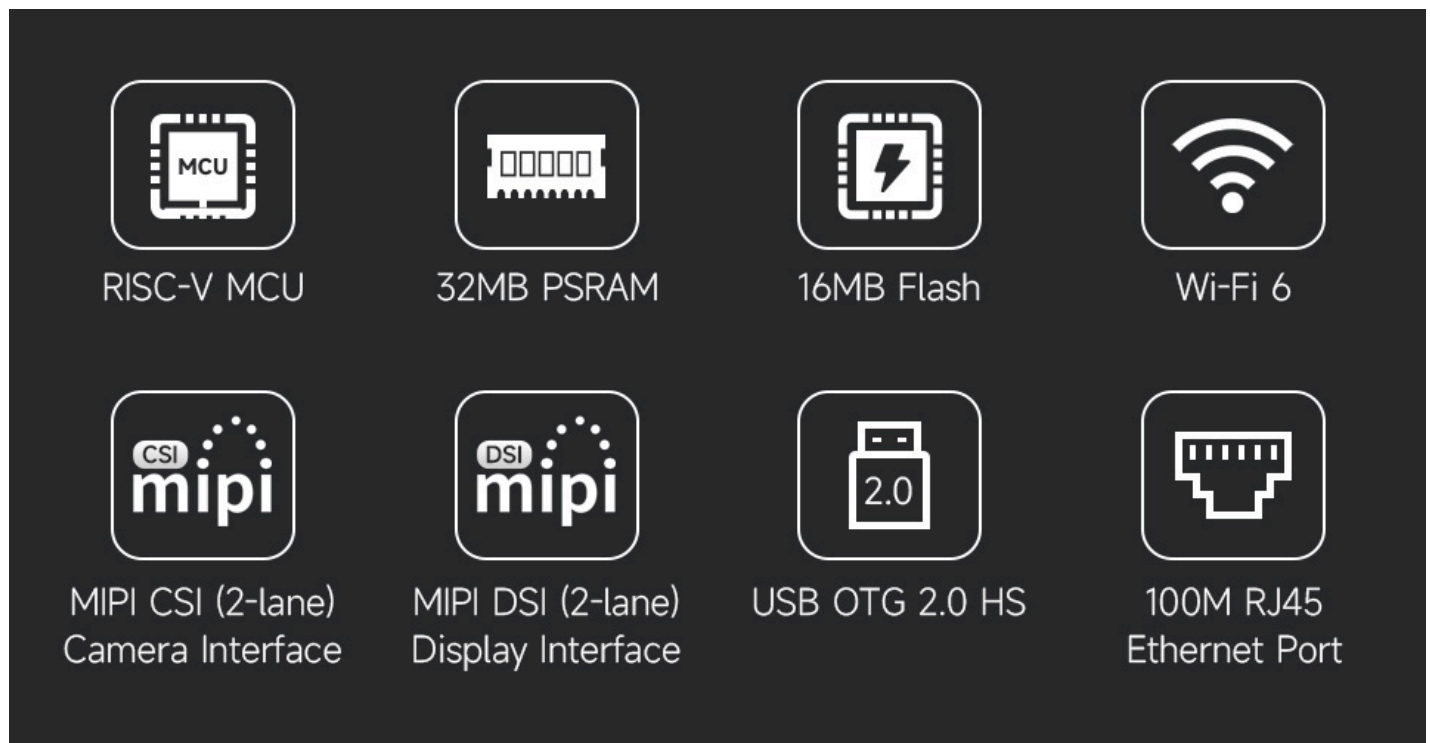
High-Performance Development Board Based On ESP32-P4 And ESP32-C6

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Rich Human-Machine Interfaces



Chat



Development Board Introduction

This product is a high-performance development board based on ESP32-P4 chip, with integrated RISC-V 32-bit dual-core and single-core processors. It features rich Human-Machine interfaces, including MIPI-CSI (with integrated Image Signal Processor), MIPI-DSI, SPI, I2S, I2C, LED PWM, MCPWM, RMT, ADC, UART,

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TWAI, etc. Additionally, it supports USB OTG 2.0 HS, Ethernet port and SDIO Host 3.0 for high-speed connectivity.

The ESP32-P4 chip integrates the Digital Signature Peripheral and a dedicated Key Management Unit, ensuring secure data and operations. Specifically designed for high-performance and high-security applications, the ESP32-P4-WIFI6-DEV-KIT meets the requirements of Human-Machine interaction, efficient edge computing, and IO expansion.

Key features include:

- High-performance MCU with RISC-V 32-bit dual-core and single-core processors
- 128 KB HP ROM, 16 KB LP ROM, 768 KB HP L2MEM, 32 KB LP SRAM, 8 KB TCM
- Powerful image and voice processing capability. Provides image and voice processing interfaces including JPEG Codec, Pixel Processing

Accelerator, Image Signal Processor, H264 encoder

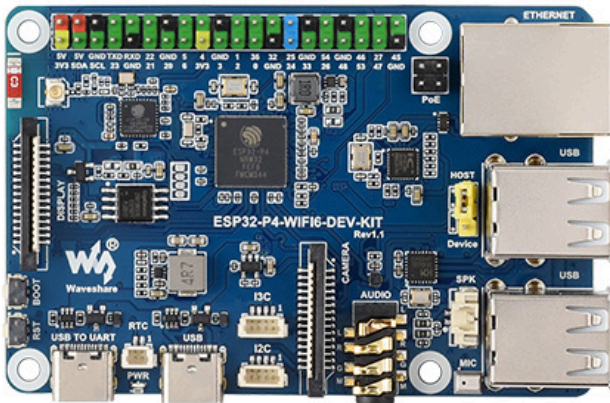
- 32MB PSRAM in the chip's package, with onboard 16MB Nor Flash
- Commonly used peripherals such as MIPI-CSI, MIPI-DSI, USB 2.0 OTG, Ethernet, SDIO 3.0 TF card slot, microphone, speaker header and RTC battery header, etc.
- Adapting 2*20 GPIO headers with 28 x remaining programmable GPIOs
- Security features: Secure Boot, Flash Encryption, cryptographic accelerators, and TRNG.

Additionally, hardware access protection mechanisms help to enable Access Permission Management and Privilege Separation

Kit Selection

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ESP32-P4-WIFI6-DEV-KIT Basic Kit



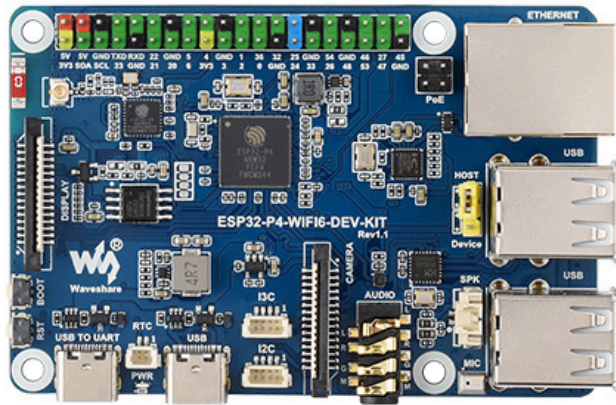
ESP32-P4-WIFI6-DEV-KIT



Speaker

ESP32-P4-WIFI6-DEV-KIT-A

ESP32-P4-WIFI6-DEV-KIT



Speaker



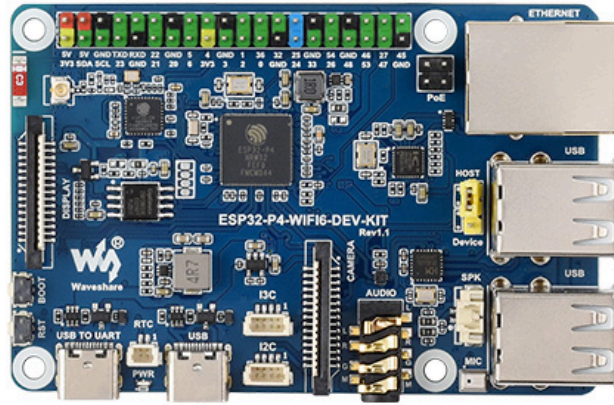
RPi Camera (B)



FFC 15PIN Cable

ESP32-P4-WIFI6-DEV-KIT-B

ESP32-P4-WIFI6-DEV-KIT



Speaker



RPi Camera (B)



FFC 15PIN Cable



7-DSI-TOUCH-A



DSI Cable



2PIN To 3PIN Cable



USB Type-A
Dual-plug Cable

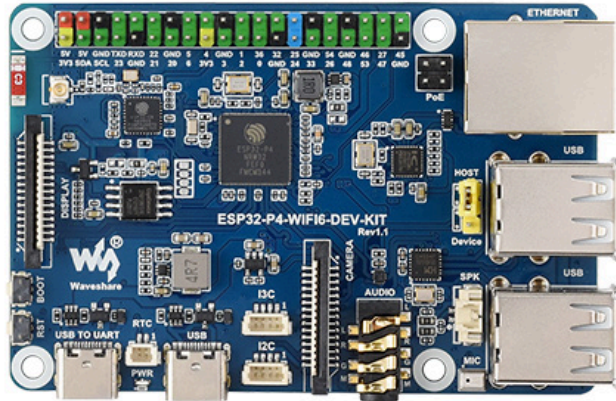


USB Type-A To
Type-C Cable

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ESP32-P4-WIFI6-DEV-KIT-C

ESP32-P4-WIFI6-DEV-KIT



Speaker



RPi Camera (B)



FFC 15PIN Cable



10.1-DSI-TOUCH-A



DSI Cable



2PIN To 3PIN Cable



USB Type-A
Dual-plug Cable



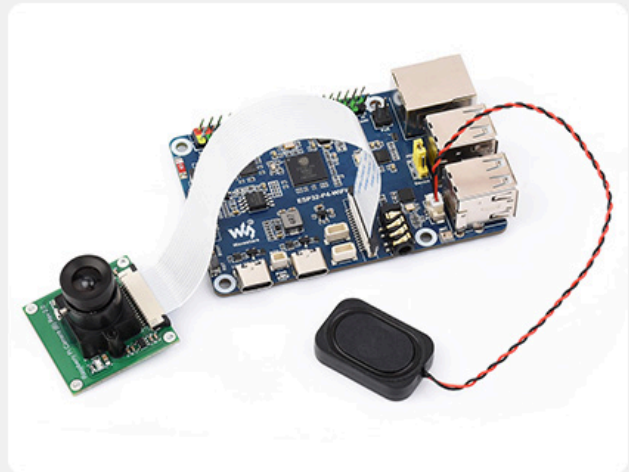
USB Type-A To
Type-C Cable

Kit Connection Examples

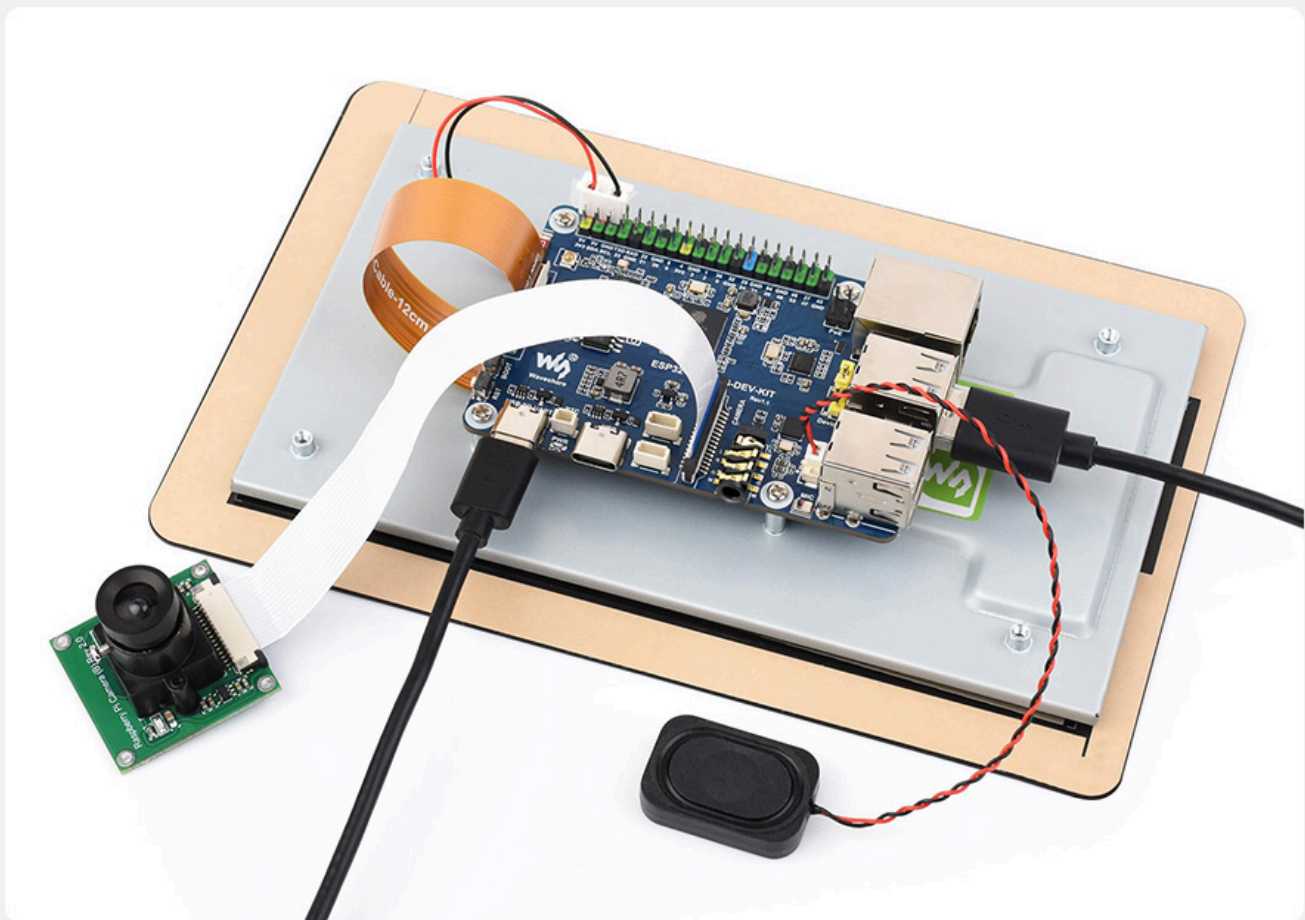
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ESP32-P4-WIFI6-DEV-KIT Basic Kit



ESP32-P4-WIFI6-DEV-KIT-A



ESP32-P4-WIFI6-DEV-KIT-B

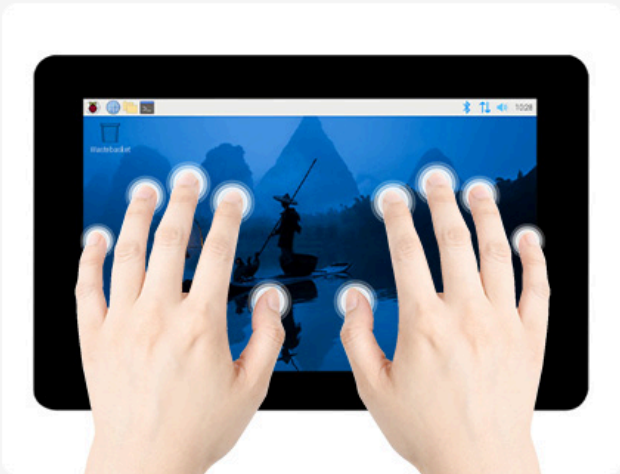
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DSI Capacitive Touch Display Comparison

<https://www.waveshare.com/esp32-p4-wifi6-dev-kit.htm?sku=32054>

Optional For DSI Capacitive Touch Display, IPS Screen, Optical Bonding Toughened Glass Panel, With 178° Wide Viewing Angle. Supports 10-Point Touch, Provides Clear Image And Excellent Visual Experience



7-DSI-TOUCH-A



10.1-DSI-TOUCH-A

MODEL	7-DSI-TOUCH-A	10.1-DSI-TOUCH-A
RESOLUTION	720 × 1280 (H×V)	800 × 1280 (H×V)
COMMUNICATION INTERFACE	MIPI 2-lane	

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BRIGHTNESS	450cd/m ² (supports customization for 1000cd/m ² high brightness version)	400cd/m ² (supports customization for 1000cd/m ² high brightness version)
TOUCH CHIP	GT9271	
OUTLINE DIMENSION	120.24 × 189.32 (mm)	147.0 × 239.0 (mm)
DISPLAY AREA	88.0 × 155.0 (mm)	135.36 × 216.58 (mm)
PIXEL PITCH	0.12075 (H) × 0.12075 (V) (mm)	0.1692 (H) × 0.1692 (V) (mm)
CONTRAST RATIO	800:1	
OPERATION TEMPERATURE	0°C~60°C	

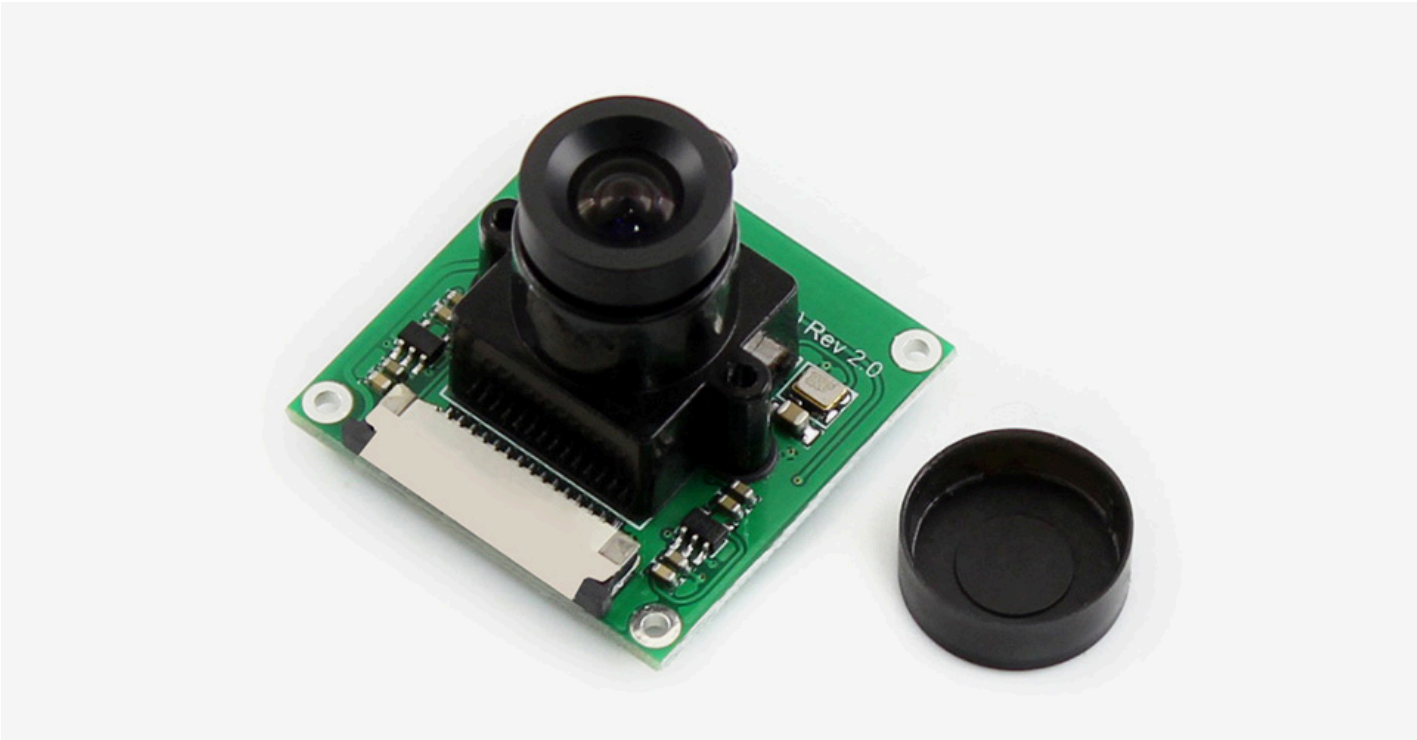
* the DSI LCD is only included in the ESP32-P4-WIFI6-DEV-KIT-B and ESP32-P4-WIFI6-DEV-KIT-C.

Raspberry Pi Camera (B)

[Chat](#)

5MP | Manual Focus

Compatible With Raspberry Pi Series Boards



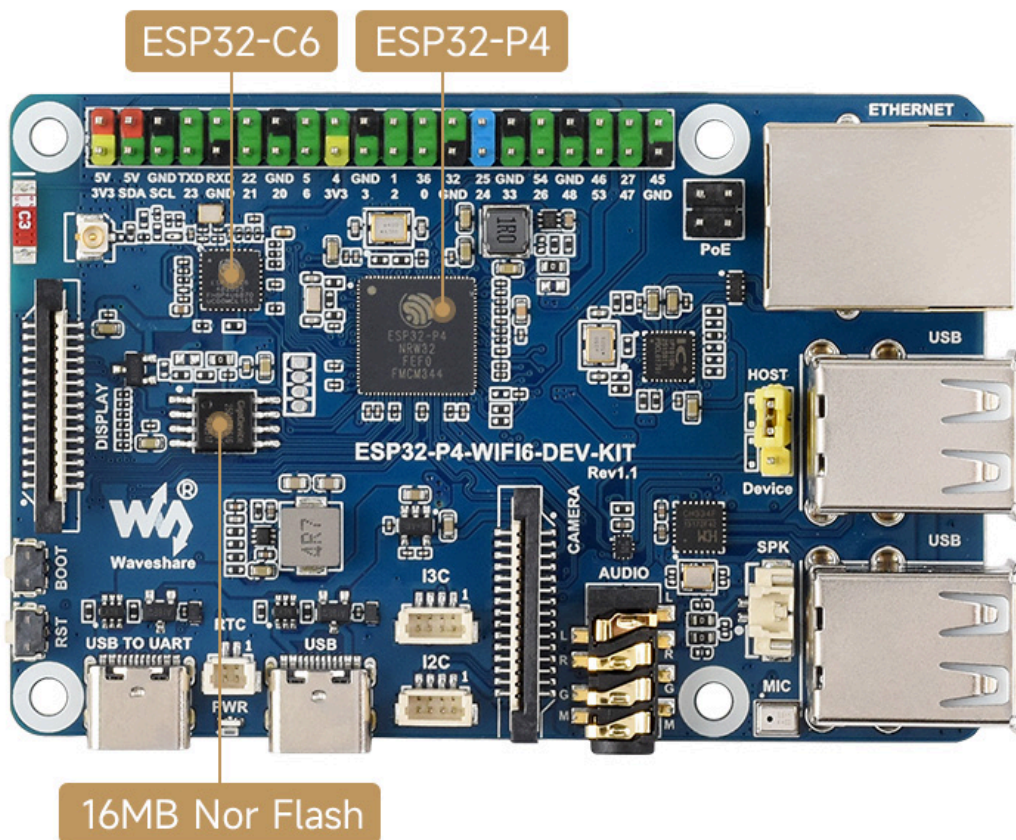
SENSOR	OV5647	CMOS SIZE	1/4inch
PIXELS	5MP	APERTURE	F2.0
FOCAL LENGTH	6mm	FIELD OF VIEW	60.6°
NIGHT VISION FUNCTION	Not support	FOCUSING	Manual Focus

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Highly Integrated And Powerful Performance

Adopts ESP32-P4 Module, Onboard ESP32-C6 And 16MB Nor Flash

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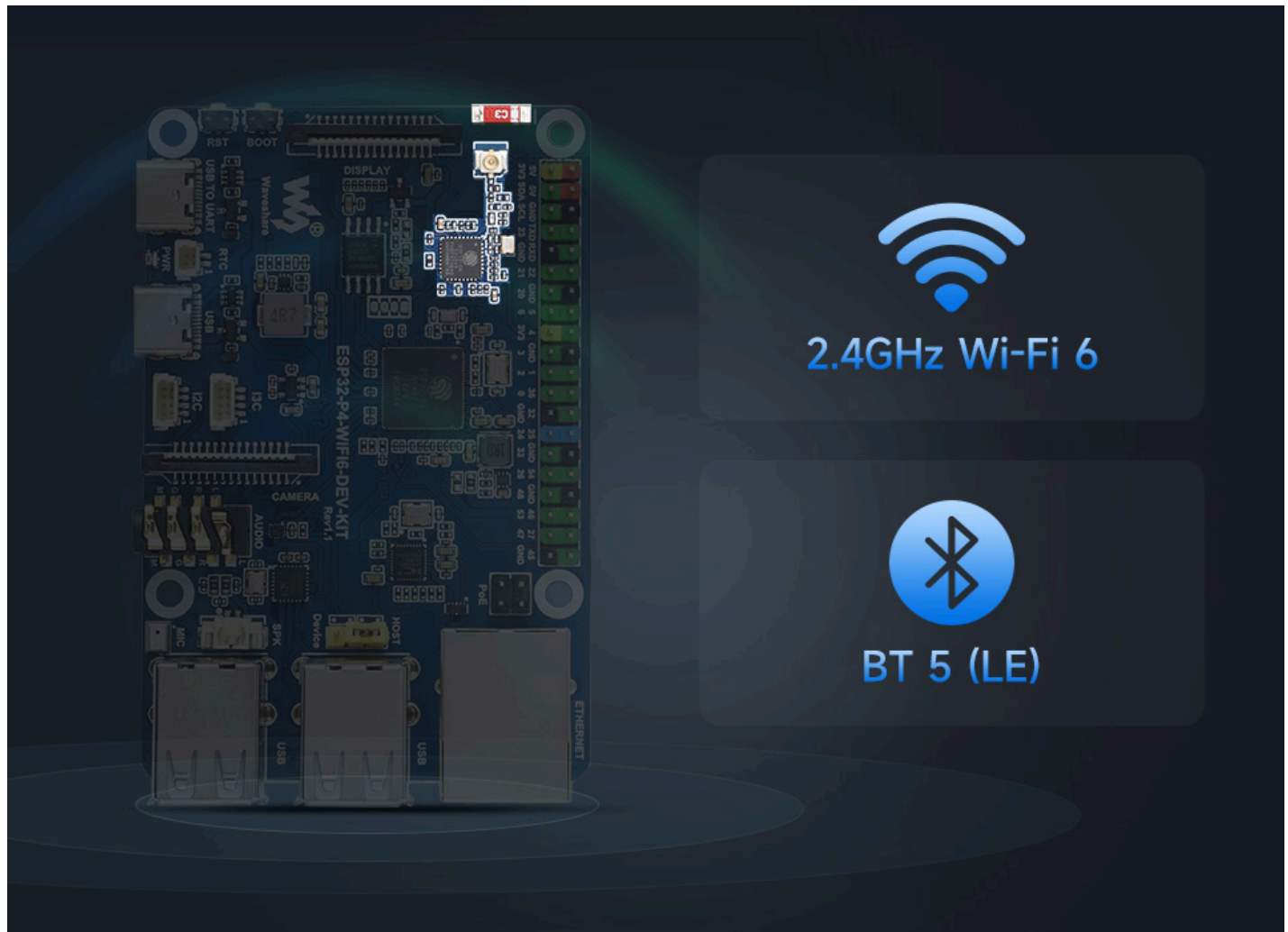
WiFi 6 And Bluetooth Module

Onboard ESP32-C6 Chip To Extend 2.4GHz Wi-Fi 6 And Bluetooth 5/BLE

For ESP32-P4-WIFI6-DEV-KIT, Using SDIO Interface Protocol

Chat

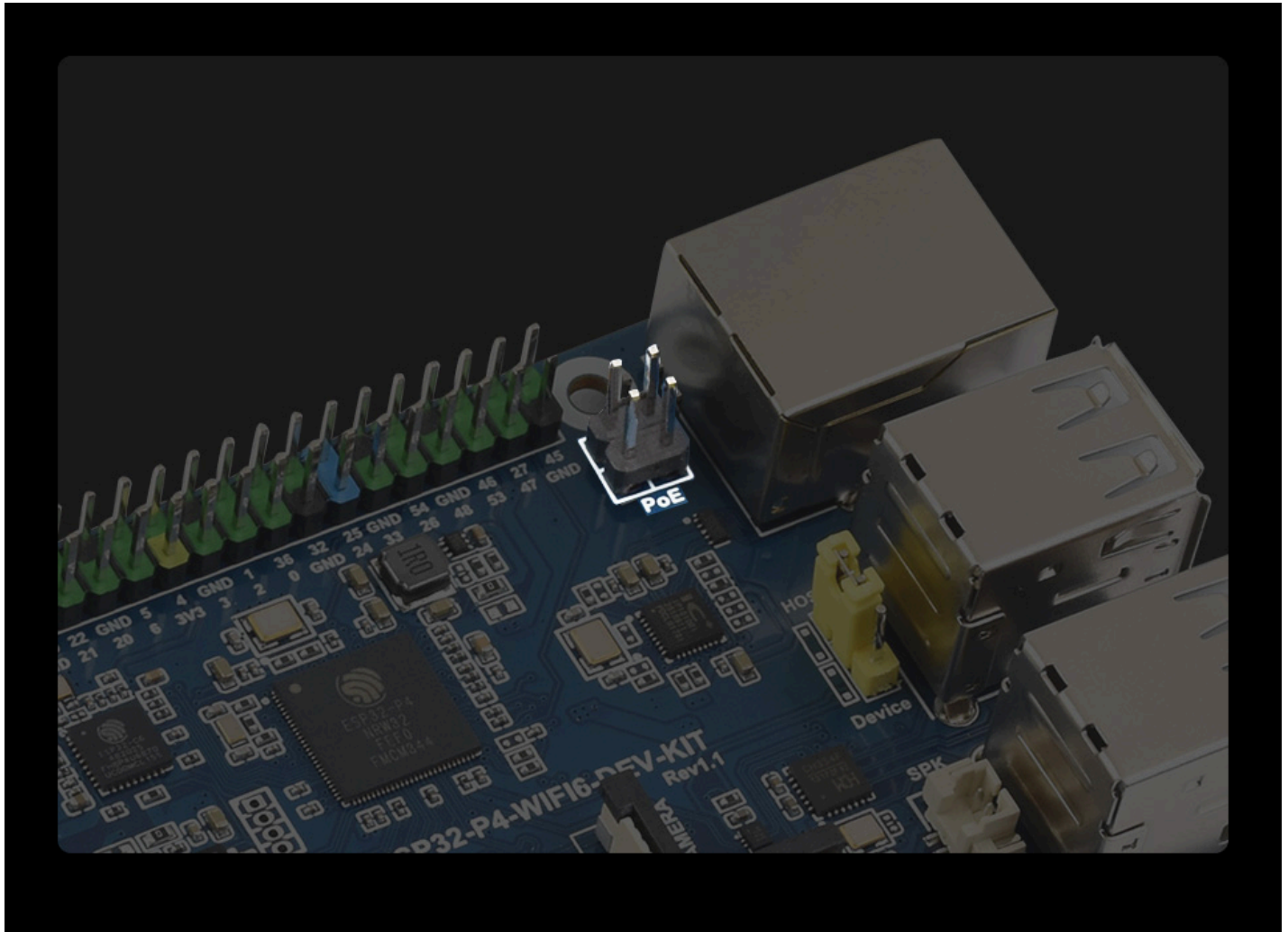
For Communication,
Stable Connection And Efficient Transmission



Reserved PoE Module Header

More Flexible For Power Supply

Chat

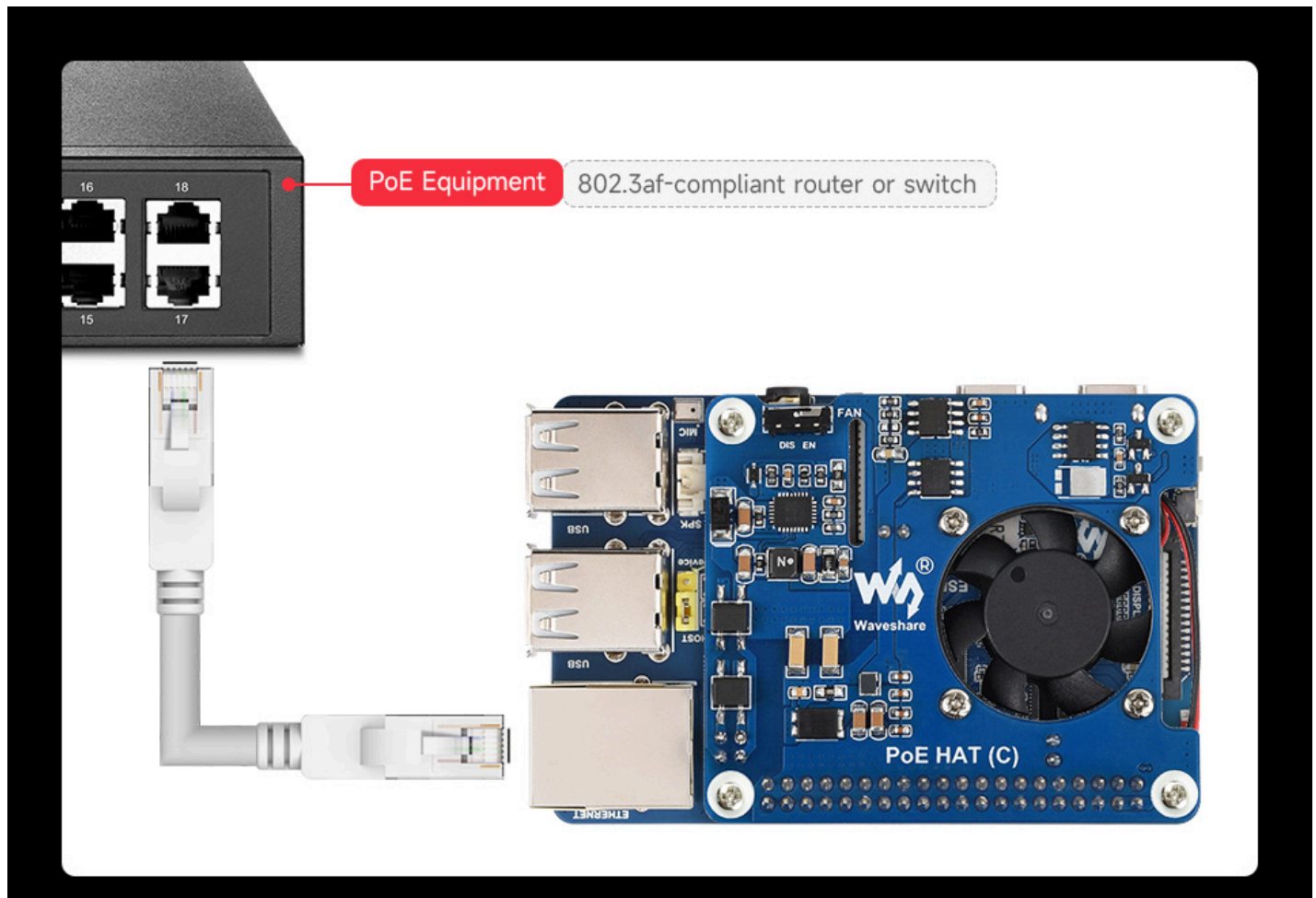


Connect To A PoE Module For PoE Power Supply

Provides Both Network Connection And Power Supply For ESP32-P4-WIFI6-DEV-KIT Board With Only One Ethernet

Chat

Cable



* the [PoE HAT \(C\)](#) and the switch in the picture above are NOT included.

Supports AI Speech Interaction

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Allows Access To Online Large Model Platforms Such As
DeepSeek, Doubao, Etc.

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Send to large model
platform for analysis

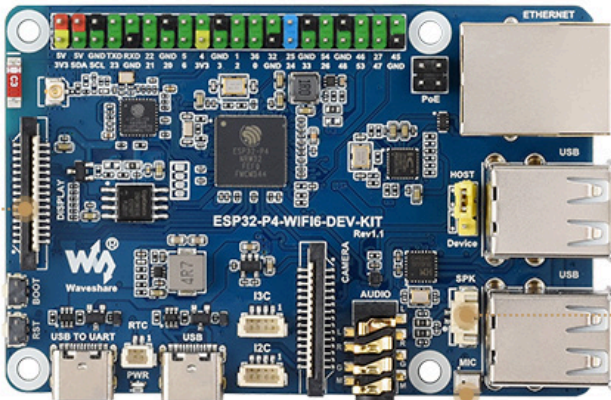


Get the answer

Display output (connect to a DSI display)



Get the reply
(connect to a speaker)

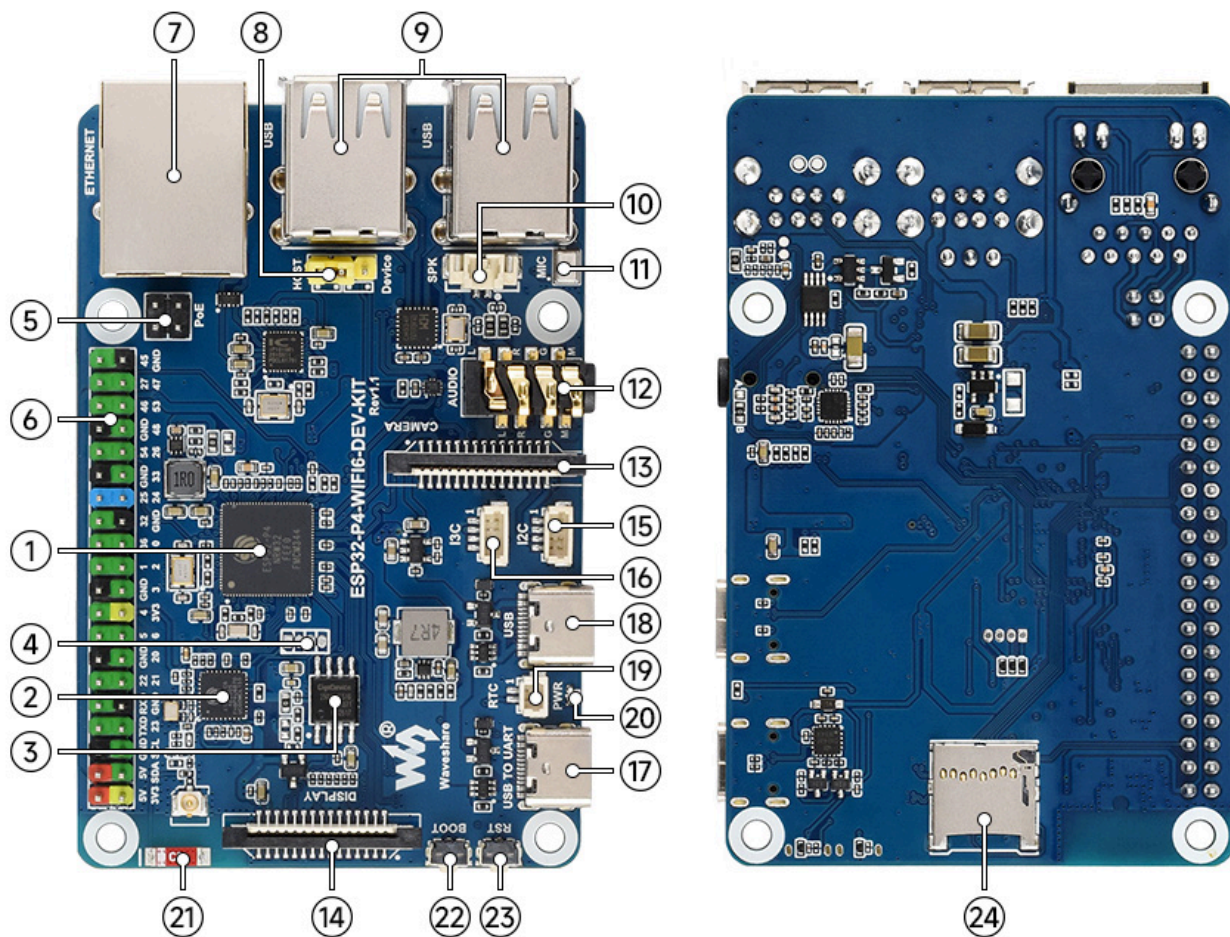


Make a conversation (with built-in microphone)

Chat

* The DSI LCD is only included in the ESP32-P4-WIFI6-DEV-KIT-B/C

What's On Board



Chat

1. ESP32-P4

ESP32-P4 chip, with 32MB PSRAM

2. ESP32-C6

SDIO 3.0 interface protocol, extending Wi-Fi 6 / Bluetooth 5 (LE) for ESP32-P4

3. 16MB Nor Flash**4. ESP32-C6 UART pads****5. PoE module header**

connect to external PoE module (NOT included) for PoE power supply function

6. 40PIN GPIO header**7. 100M RJ45 Ethernet port****8. USB OTG function selection**

switching to HOST end for expanding USB ports

9. USB-A ports

USB OTG 2.0 high-speed ports, switching to HOST or DEVICE via jumper

10. Speaker header

MX1.25 2P connector, supports 8Ω 2W speaker

11. Onboard microphone**12. 3.5mm audio jack**

Supports audio playback via connected headphones

13. Camera interface

MIPI-CSI (2-lane), supports [OV5647](#) camera, etc

14. Display interface

MIPI-DSI (2-lane), supports [5](#) / [7](#) / [8](#) / [10.1](#) inch DSI screen

15. I2C interface**16. I3C interface****17. Type-C UART port**

for power supply, program burning and debugging

18. Type-C USB port

for power supply, program burning

19. RTC battery header

for connecting rechargeable [RTC battery](#) (supports rechargeable RTC batteries only)

20. PWR

Power indicator

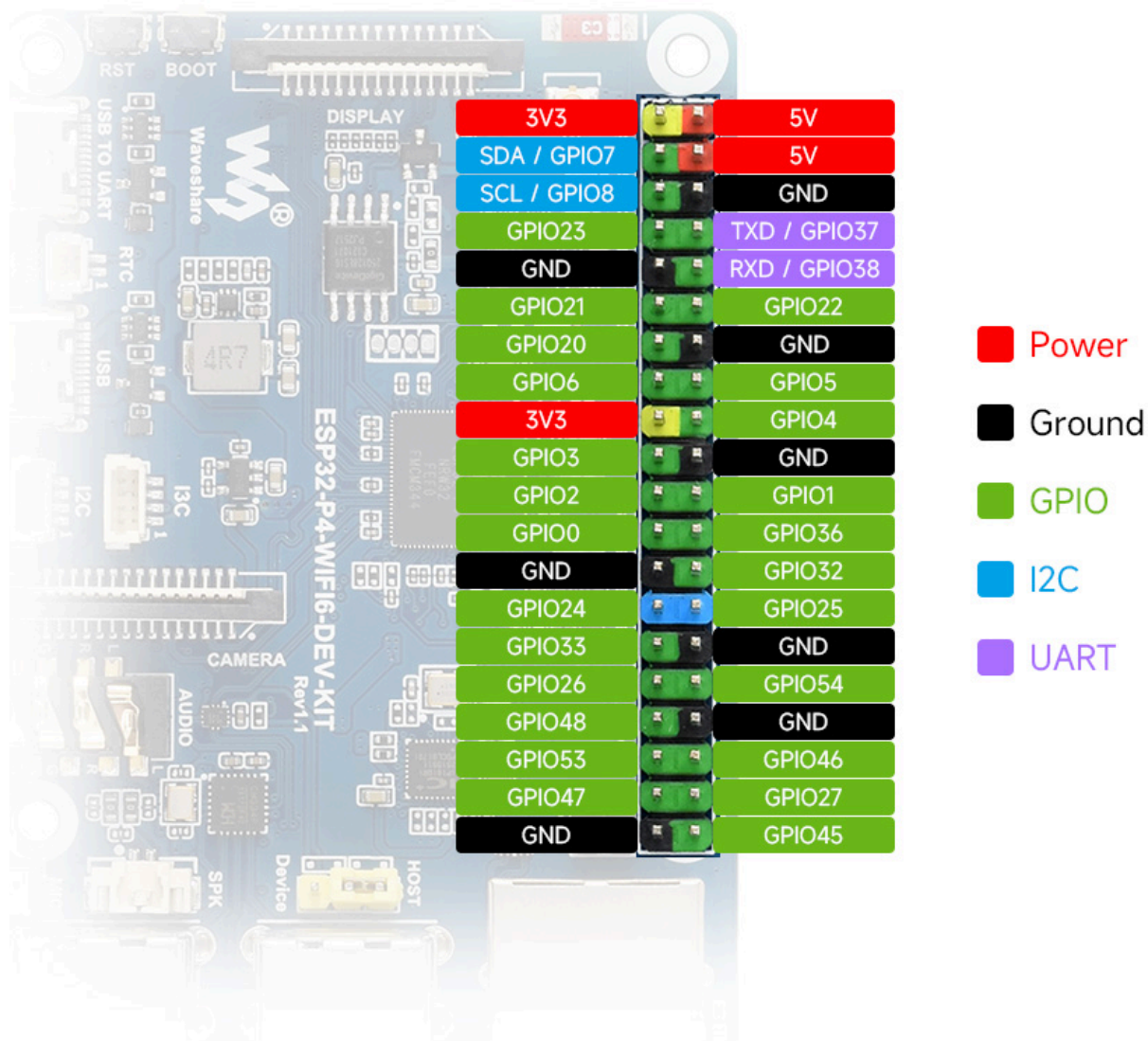
21. ESP32-C6 SMD antenna**22. BOOT Button**

Press it when powering on or resetting to enter download mode

23. RST Reset button**24. TF card slot**

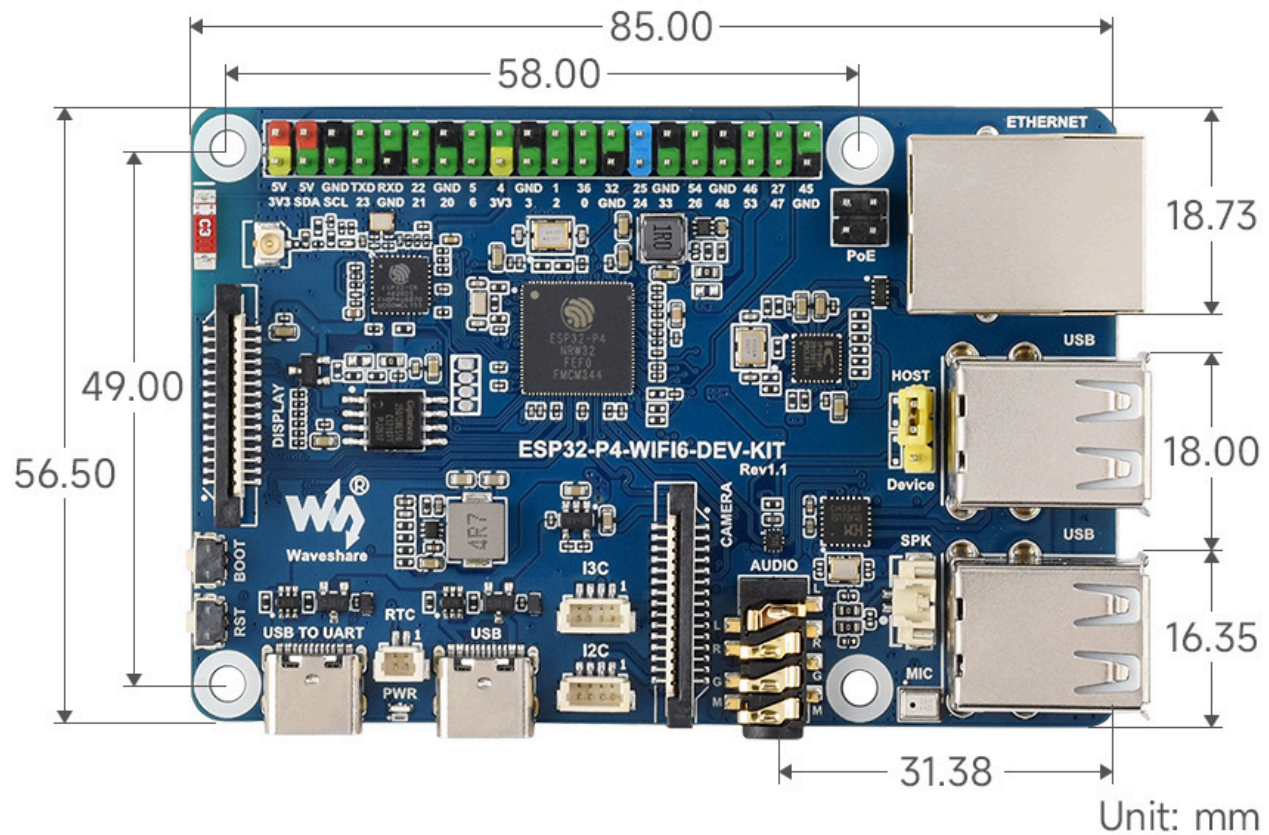
SDIO 3.0 interface protocol

Pin Definition



Outline Dimensions

[Chat](#)



Resources & Services



WAVESHARE
ORIGINAL ✓



PROFESSIONAL
TECH SUPPORT



RICH RESOURCES
MANUAL & DEMO

Wiki: docs.waveshare.com/ESP32-P4-WIFI6-DEV-KIT

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